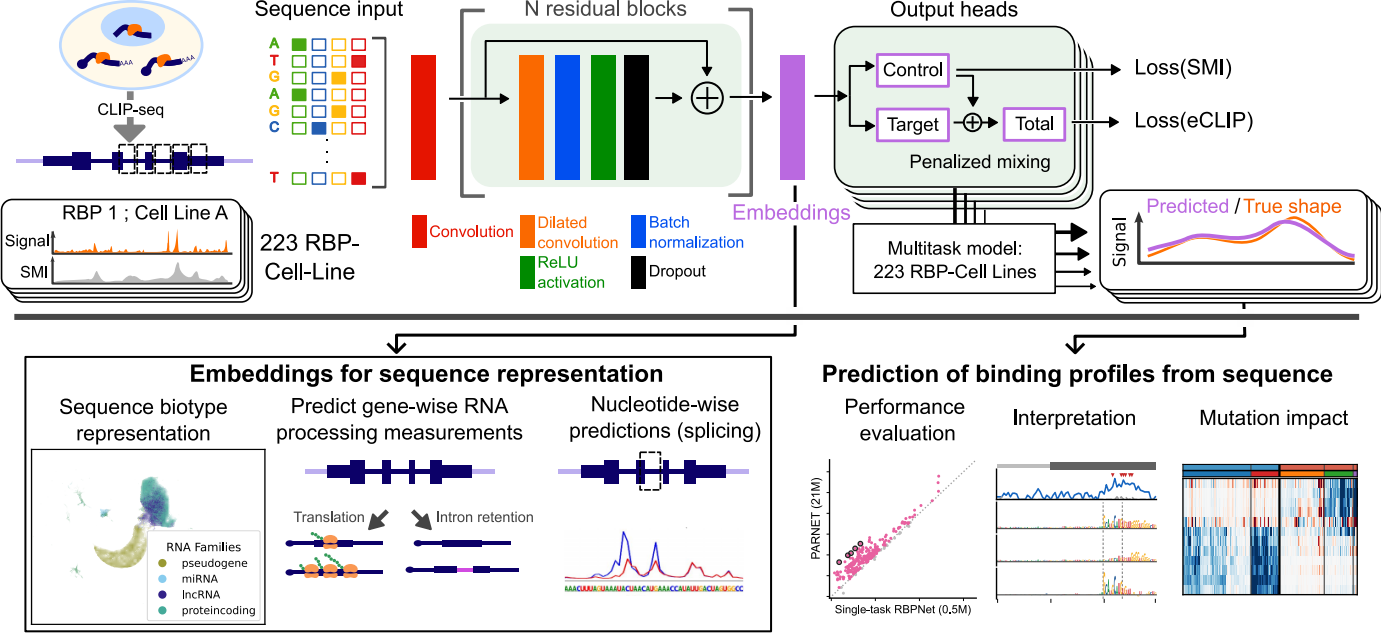

CLIP-seq
RBP 1 ; Cell Line A
Signal
SMI
223 RBP-Cell-Line
Sequence input
A T G A G C T
N residual blocks
Convolution
Dilated convolution
Batch normalization
ReLU activation
Dropout
Embeddings
Output heads
Control
Target
Total
Penalized mixing
Loss(SMI)
Loss(eCLIP)
Multitask model: 223 RBP-Cell Lines
Predicted / True shape
Signal
Embeddings for sequence representation
Sequence biotype representation
RNA Families
pseudogene
miRNA
lncRNA
proteincoding
Predict gene-wise RNA processing measurements
Translation
Intron retention
Nucleotide-wise predictions (splicing)
Prediction of binding profiles from sequence
Performance evaluation
PARNET (21M)
Single-task RBPNet (0.5M)
Interpretation
Mutation impact